Hijacking 3D Spatial Reasoning: Query-Conditioned Relational Adversarial Attacks on Point-Cloud Vision-Language Models

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Top-Conference Style Research Manuscript Draft

Abstract

3D vision-language models (3D VLMs) are moving beyond object-level point-cloud recognition toward scene-level question answering, natural-language grounding, relation-aware captioning, and embodied decision making. This shift creates a security problem that is not captured by classical 3D adversarial benchmarks: a model may correctly recognize all relevant objects while giving an incorrect answer about their spatial relation. Such relation-level failures are subtle, because the semantic layer appears intact, yet safety-critical decisions such as target selection, manipulation, and navigation can be corrupted. We introduce RelAttack3D, a query-conditioned relational adversarial attack for point-cloud VLMs. Given a point-cloud scene and a natural-language query, RelAttack3Dparses the query into a relation graph, localizes the involved objects, mines relation-critical points, and optimizes constrained perturbations that induce relation-level errors while preserving object semantics, local geometry, point density, and sensor plausibility. Unlike classification-critical point attacks, RelAttack3Dexplicitly models relation-dependent geometric evidence, including centroid projections, support/contact surfaces, nearest inter-object point pairs, containment boundaries, and free-space corridor margins. We further propose RelBench-3D, a relation-robustness evaluation protocol spanning 3D visual question answering, natural-language grounding, relation-aware captioning, and embodied affordance judgment. Finally, we introduce Relation-Consistency Verification, a lightweight detection baseline based on logical query rewriting, explicit geometric checking, and viewpoint consistency. This manuscript provides a detailed top-conference-style formulation, method, benchmark construction procedure, implementation plan, ablation design, and result-table templates. Numerical entries are intentionally left as placeholders and must be replaced by real experiments before submission.

1 Introduction

3D vision-language models (3D VLMs) are becoming a natural interface between geometric perception and language-driven reasoning. Early point-cloud learning focused on shape classification, part segmentation, semantic segmentation, object detection, registration, completion, and reconstruction. Recent 3D VLMs instead ask a more ambitious question: can a model use point clouds as evidence for open-ended language interaction? PointLLM connects point-cloud encoders with large language models for object-level point-language instruction following [5]; GPT4Point presents a unified point-language framework for understanding and generation [6]; and 3D-LLM injects 3D world representations into large language models to support 3D question answering, dense captioning, grounding, dialogue, and navigation [7]. These models suggest that point clouds are no longer merely inputs to discriminative geometric networks. They increasingly serve as multimodal evidence for language-conditioned spatial reasoning.

This change forces a change in how robustness should be defined. In classical point-cloud adversarial learning, the dominant attack goal is object-level misclassification: a perturbed airplane is classified as a chair, or a perturbed chair is classified as a table. This threat model remains relevant, but it is incomplete for 3D VLMs. A 3D VLM may correctly recognize a chair and a table, produce a fluent description, and still answer incorrectly when asked whether the chair is to the left of the table. It may recognize a sofa and a wall, yet incorrectly conclude that a robot can pass between them. It may identify a cup and a desk, but fail to judge whether the cup is on the desk or under it. These failures are not merely semantic recognition errors; they are *relation-level reasoning failures*.

Relation-level failures are especially dangerous because spatial relations are the bridge between perception and action. Directional relations such as left/right and front/behind support target search and navigation. Support relations such as on/under support manipulation. Distance and comparative relations support object disambiguation. Topological and affordance relations such as between, blocked, passable, reachable, and inside support embodied decision making. A system that preserves object recognition but corrupts these relations may appear reliable to a human evaluator while still causing downstream planning or interaction failures.

Existing 3D adversarial attacks have shown that point-cloud classifiers and perception systems are vulnerable to point perturbation, point addition, point dropping, and minimal adversarial modifications [12, 13, 14]. Emerging work has also begun to study the adversarial robustness of 3D large VLMs and shows that point-based 3D VLMs can be vulnerable to vision-token and caption-level attacks [16]. However, these studies do not directly address a key question:

> Can an adversary preserve the semantic identity of the relevant objects and the plausibility of the point cloud while selectively hijacking the spatial relation reasoning of a 3D VLM under a natural-language query?

We argue that this question defines a distinct and underexplored attack surface. A successful relational attack should not simply make the model confused about what objects exist. It should keep the chair as a chair, the table as a table, and the scene visually plausible, while flipping or corrupting the answer to a relation query. This is a stronger and more diagnostic criterion than generic caption degradation.

Key observation. The points that determine object identity are not necessarily the points that determine a spatial relation. For recognizing a chair, the chair back, seat, armrests, and legs may be discriminative. For answering whether the chair is left of a table, however, relation-relevant evidence may be concentrated around projected centroids, lateral boundaries, nearest inter-object surfaces, or viewpoint-dependent bounding-box order. For answering whether a robot can pass between a sofa and a wall, the relevant points are not the most class-discriminative sofa points, but the clearance boundary and free-space corridor margin. Thus, attacking a 3D VLM's relation reasoning requires a query-conditioned and relation-aware perturbation strategy.

We propose RelAttack3D, a query-conditioned relational adversarial attack framework for point-cloud VLMs. Given a point cloud $\mathcal{P}$ and a natural-language query q, RelAttack3Dfirst parses the query into a structured relation graph. It then localizes the involved objects and computes a relation-critical score for each point by combining model sensitivity, relation-specific geometric support, boundary importance, and reference-frame consistency. Only the top-ranked relation-critical points are perturbed. The perturbation is optimized to induce a target relation error while preserving object semantics, global geometry, local smoothness, point density, and sensor plausibility.

The goal of this paper is not to claim a minor improvement over existing point-cloud attacks. The goal is to establish a more precise robustness problem for 3D VLMs: *semantic-preserving relation hijacking*. If a small, plausible point-cloud perturbation can systematically change relation answers while leaving object recognition intact, then object-level robustness and caption-level robustness are insufficient proxies for 3D VLM safety.

Contributions. This manuscript makes the following contributions:

- We formulate **semantic-preserving relational attack** as a new robustness problem for 3D VLMs. A valid attack must corrupt a relation answer while preserving the identities of the queried objects and satisfying geometric and sensor-plausibility constraints.
- We introduce **query-conditioned relation-critical point mining**, which selects points according to language-conditioned relation evidence rather than classification saliency alone.
- We design **relation-specific attack objectives** for 3D VQA, natural-language grounding, relation-aware captioning, and embodied affordance judgment.
- We propose **RelBench-3D**, a benchmark protocol that extracts relation-centric samples from ScanQA, ScanRefer, SQA3D, ReferIt3D-style referring expressions, and embodied scene questions.
- We provide a lightweight defense baseline, **Relation-Consistency Verification**, which checks logical equivalence, geometric relation consistency, and viewpoint transformation consistency.
- We provide detailed implementation, evaluation, ablation, and reporting templates suitable for converting this draft into a reproducible conference submission.

2 Related Work

2.1 Point-Cloud Representation Learning

Deep learning on point clouds has progressed from simple permutation-invariant architectures to hierarchical, graph-based, transformer-based, and masked-pretraining models. PointNet [1] introduced a symmetric aggregation mechanism for unordered point sets. PointNet++ [2] added hierarchical local feature learning. DGCNN [3] used dynamic graphs to capture local geometric neighborhoods, while Point-BERT [4] explored masked point modeling for 3D representation pre-training. These models remain important as point encoders for downstream 3D-language systems.

However, representation learning alone does not solve relation reasoning. A point encoder may recognize object geometry, but relation reasoning requires object localization, reference-frame interpretation, relational comparison, and sometimes physical affordance reasoning. Our work studies failures at this relation layer.

2.2 3D Vision-Language Models

3D VLMs connect point-cloud perception with natural-language understanding. PointLLM [5] aligns point-cloud features with a large language model and supports generative 3D object classification and captioning. GPT4Point [6] proposes a unified framework for point-language understanding and generation. 3D-LLM [7] extends large language models into 3D scenes and supports 3D captioning, question answering, grounding, task decomposition, and navigation.

These systems demonstrate a transition from object-level point-language alignment to scene-level reasoning. As 3D VLMs are increasingly used for grounded interaction, robustness must be evaluated not only by whether objects are named

correctly, but also by whether spatial relations and action-relevant properties remain reliable.

2.3 3D Spatial Question Answering and Grounding

ScanRefer [8] introduced natural-language 3D object localization in RGB-D scans. ScanQA [9] formulated 3D question answering for spatial scene understanding. SQA3D [10] considered situated question answering, where the agent's position and orientation affect the answer. ReferIt3D-style datasets [11] contain fine-grained referring expressions that often depend on relational object disambiguation.

These benchmarks are useful foundations for relational robustness, but they were not designed as adversarial robustness benchmarks. They typically evaluate clean-input task accuracy rather than whether relation answers are stable under small, structured, query-conditioned point-cloud perturbations.

2.4 Adversarial Attacks on 3D Point Clouds

Xiang et al. [12] introduced adversarial point-cloud generation and perturbation attacks. Later work studied minimal adversarial point modifications [13], transferability, outlier-aware constraints, robust training, and defense. Surveys summarize attack types such as point shifting, point dropping, point adding, transformation attacks, and optimization-based attacks [14].

Most of these attacks target classifiers or perception modules. Their success criterion is usually a wrong class label or wrong detection result. By contrast, RelAttack3Dtargets language-conditioned relation answers while explicitly preserving the semantic identities of the queried objects. This distinction matters because an attack that changes "chair" into "table" does not prove that relation reasoning is vulnerable; it only proves that object recognition is vulnerable.

2.5 Adversarial Robustness of Vision-Language Models

Large VLMs are known to be vulnerable to adversarial visual perturbations, and robustness evaluation has become an important topic in 2D multimodal learning [15]. Recent work on 3D large VLM robustness studies adversarial vulnerability in point-based 3D VLMs and introduces vision attacks and caption attacks [16]. This confirms that adversarial evaluation should be extended to 3D multimodal models.

Our work differs by focusing on a structured failure mode: relation hijacking. Instead of simply degrading captions or forcing arbitrary target text, we ask whether a model can be made wrong about a specific relation while still recognizing the involved objects. This makes the attack more diagnostic for embodied and safety-critical settings.

3 Problem Formulation

3.1 3D VLM Inference

Let a point-cloud scene be

$$\mathcal{P} = \{p_i\}_{i=1}^{N}, \quad p_i = (x_i, y_i, z_i, c_i), \tag{1}$$

where $(x_i, y_i, z_i) \in \mathbb{R}^3$ are coordinates and c_i denotes optional color, intensity, semantic feature, or learned embedding. A 3D VLM F_θ receives $\mathcal{P}$ and a natural-language query q and produces an output

$$F_\theta(\mathcal{P}, q) \to y. \tag{2}$$

The output y may be a short answer, a free-form caption, a grounding box, a segmentation mask, or an action recommendation.

We focus on queries that contain one or more spatial relations. A query parser maps q into a relation graph

$$\mathcal{G}_q = (\mathcal{V}_q, \mathcal{E}_q), \tag{3}$$

where each node is an object mention and each edge is a spatial relation. A binary relation edge is written as

$$\tau = (o_s, r, o_t, \pi), \tag{4}$$

where o_s is the subject object, o_t is the reference object, r is the relation type, and π is the reference frame. Comparative relations can involve multiple references, e.g.

$$\tau = (o_s, \text{closer_than}, o_{t1}, o_{t2}, \pi). \tag{5}$$

3.2 Reference Frames

Directional relations are undefined without a reference frame. The phrase "left of the table" may refer to a world frame, a camera frame, an observer frame, a robot ego frame, or an object-centric frame. We therefore define a directional relation as

$$r = (r_{\text{type}}, \pi), \tag{6}$$

where π specifies the coordinate convention or pose. In datasets without reliable view or agent pose, we do not evaluate left/right and front/behind as primary relation labels. Instead, we focus on viewpoint-invariant or weakly viewpoint-dependent relations such as on/under, near/far, inside/outside, between, adjacent to, and supported by. In situated datasets, directional labels are evaluated under the provided pose description.

This design prevents a common benchmark flaw: measuring attack success on a relation whose ground truth is ambiguous. A top-conference submission must be explicit about this issue; otherwise the attack may be rejected as ill-posed.

3.3 Relation Taxonomy

Table 1 summarizes the relation families considered in this work. Each family has different geometric evidence and therefore a different attack surface.

Table 1: Relation taxonomy used by RelAttack3D. The key point is that each relation family has different geometric evidence and should not be attacked with a single generic point saliency map.

Relation Family	Examples	Dominant Geometric Evidence	Relation-Critical Attack Surface
Directional	left/right, front/behind	Relative centroid projection and projected bounding-box order under frame π	Lateral or frontal object boundaries; centroid-projection region
Contact/support	on, under, above, below	Vertical ordering, support/contact plane, surface normals	Contact surface, lower/upper boundary, local normal field
Distance	near, far, closest to	Euclidean or geodesic distance between object surfaces or centers	Nearest inter-object point pairs; facing boundaries
Containment	inside, outside, surrounded by	Inclusion relation between object and container volume	Container boundary shell, opening region, enclosed object boundary
Comparative	closer than, higher than, larger than	Ordered comparison between multiple references or attributes	Points affecting relative order or measured attribute
Topological	between, next to, adjacent to	Scene adjacency graph, object layout, free-space region	Corridor boundary, obstacle margin, inter-object middle region
Affordance/safety	passable, blocked, reachable	Traversability, robot footprint clearance, collision margin	Narrow passage, obstacle boundary, clearance bottleneck

3.4 Valid Relational Attack

A conventional point-cloud attack is successful when the predicted class changes. A relational attack requires a stricter criterion. Given a clean scene $\mathcal{P}$, query q, clean answer y, target wrong answer y_{tar}, and adversarial scene $\mathcal{P}^{\text{adv}}$, a sample is a *valid relational attack* if it satisfies four conditions:

$$\textbf{Relation error:} \quad g(F_\theta(\mathcal{P}^{\text{adv}}, q)) = y_{tar} \quad \text{or} \quad g(F_\theta(\mathcal{P}^{\text{adv}}, q)) \neq g(F_\theta(\mathcal{P}, q)), \tag{7}$$

$$\textbf{Semantic preservation:} \quad h(o \mid \mathcal{P}^{\text{adv}}) = h(o \mid \mathcal{P}), \quad \forall o \in \mathcal{O}_q, \tag{8}$$

$$\textbf{Geometric plausibility:} \quad D_{geo}(\mathcal{P}, \mathcal{P}^{\text{adv}}) \leq \varepsilon_g, \tag{9}$$

$$\textbf{Sensor plausibility:} \quad \mathcal{P}^{\text{adv}} \in \mathcal{S}_{sensor}. \tag{10}$$

where g maps a free-form answer to a normalized relation label, h is an object-recognition or object-grounding validator, $\mathcal{O}_q$ is the set of objects involved in the query, D_{geo} is a geometric distance, and $\mathcal{S}_{sensor}$ is a set of sensor-plausible point clouds.

The semantic preservation condition is crucial. Without it, an attack can trivially flip a relation answer by destroying object identity. Such a failure is not evidence of relation vulnerability.

3.5 Threat Models

We consider three threat models.

White-box attack. The attacker knows the model architecture, parameters, and differentiable output distribution. This setting measures worst-case vulnerability.

Transfer attack. The attacker has access to a surrogate model F_ϕ and transfers adversarial point clouds to a target model F_θ. This setting is realistic because many deployed VLMs are inaccessible but architecturally similar.

Query-only black-box attack. The attacker can only query the target model and observe outputs. This setting uses score estimation or discrete success feedback. We treat this as the most deployment-relevant setting, but also the most expensive.

4 Method

4.1 Overview

Figure 1 gives the conceptual pipeline. RelAttack3Dtakes a point cloud and a natural-language query as input. It parses the query into relation tuples, localizes the relevant objects, computes relation-critical point scores, and optimizes perturbations only on selected points. The optimization objective combines a relation attack loss with semantic, geometric, density, and sensor-plausibility regularizers.

4.2 Query-to-Relation Graph Parsing

The parser maps a natural-language query into a relation graph $\mathcal{G}_q$. We use a hybrid design:

- A rule-based parser handles high-frequency patterns such as "A left of B", "A on B", "A closer to B than C", and "between A and B".
- A language-model parser handles compositional or paraphrased queries and returns a structured JSON-like repre-

Placeholder for Figure 1: Overview of RelAttack3D.
Input point cloud and query → relation graph parser → object localization → relation-critical point mining → constrained attack optimization → relation error with semantic preservation.

Figure 1: Conceptual overview of RelAttack3D. The final paper should replace this placeholder with a real pipeline figure and visual examples of relation-critical points.

sentation.

- A verifier checks whether the parsed relation is supported by the dataset annotations or geometric metadata.

For example, the query

"Can the robot pass between the sofa and the wall?"

produces

$$\tau = (\text{robot}, \text{passable_between}, \text{sofa}, \text{wall}, \pi_{ego}). \tag{11}$$

The query

"Which object is closer to the table, the chair or the cabinet?"

produces a comparative relation

$$\tau = (\text{chair}, \text{closer_than}, \text{table}, \text{cabinet}). \tag{12}$$

The parser output is not trusted blindly. If the query cannot be mapped to a supported relation type, the sample is excluded from RelBench-3D. This filtering prevents noisy prompts from corrupting the evaluation.

4.3 Object Localization

For each object mention $o \in \mathcal{O}_q$, we obtain an object region $\Omega_o \subset \mathcal{P}$. Depending on dataset and model access, Ω_o may come from one of the following:

- ground-truth instance masks or 3D bounding boxes;
- a frozen 3D detector or instance segmentor;
- the target 3D VLM's grounding output;
- a combination of detector proposals and language matching.

In the main benchmark, we recommend reporting two settings:

- **Oracle localization:** ground-truth object masks or boxes are used for relation-critical point mining. This isolates relation reasoning vulnerability from localization errors.
- **Predicted localization:** object proposals are produced by an external detector or the target model. This measures end-to-end vulnerability.

This separation is important. If the attack succeeds only because object localization fails, the claim is weaker. If it succeeds even under oracle localization, the relation reasoning vulnerability is more convincing.

4.4 Relation-Critical Point Mining

Let Ω_s and Ω_t be the point sets of the subject and reference objects. RelAttack3Dassigns each point p_i a relation-critical score

$$\begin{aligned} S(p_i; q, r) = \alpha S_{grad}(p_i; q) + \beta S_{geom}(p_i; r) \\ + \gamma S_{bdry}(p_i) + \eta S_{frame}(p_i; r, \pi). \end{aligned} \tag{13}$$

where S_{grad} measures model sensitivity, S_{geom} measures relation-specific geometric support, S_{bdry} measures boundary importance, and S_{frame} measures reference-frame relevance. The relation-critical set is

$$\mathcal{K}_{q,r} = \text{TopK}_{p_i \in \Omega_s \cup \Omega_t \cup \Omega_{ctx}} S(p_i; q, r), \tag{14}$$

where Ω_{ctx} contains contextual points such as inter-object corridor points, container boundaries, or support surfaces.

Gradient sensitivity. In the white-box setting,

$$S_{grad}(p_i; q) = \left\| \nabla_{p_i} \mathcal{L}_{rel}(F_\theta(\mathcal{P}, q), y_{tar}) \right\|_2. \tag{15}$$

For non-differentiable answer extraction, we compute gradients through token logits or through a differentiable relation classifier attached to the VLM output embedding.

Relation geometry. The geometric term depends on relation family. For directional relations,

$$S^{dir}_{geom}(p_i) = \exp\left(-\frac{|u_\pi^\top p_i - u_{mid}|}{\sigma_u} \right), \tag{16}$$

where u_π is the axis associated with the reference frame and u_{mid} is the midpoint of the projected object centroids. This prioritizes points near the projection boundary that can alter the model's perceived order.

For support/contact relations,

$$S^{sup}_{geom}(p_i) = \exp\left(-\frac{|z_i - z_{contact}|}{\sigma_z} \right), \tag{17}$$

where $z_{contact}$ approximates the support plane between objects.

For distance relations,

$$S^{dist}_{geom}(p_i) = \exp\left(-\frac{d(p_i, \Pi_{near})}{\sigma_d} \right), \tag{18}$$

where Π_{near} is the set of nearest inter-object point pairs.

For passability relations,

$$S^{pass}_{geom}(p_i) = \exp\left(-\frac{d(p_i, \partial\mathcal{C}_{free})}{\sigma_c}\right), \tag{19}$$

where $\partial\mathcal{C}_{free}$ is the clearance boundary of the traversable corridor.

Boundary importance. Boundary points are often more influential than interior points for relation judgments. We define

$$S_{bdry}(p_i) = 1 - \frac{d(p_i, \partial\Omega_o)}{\max_{p_j\in\Omega_o} d(p_j, \partial\Omega_o) + \xi}, \tag{20}$$

where $\partial\Omega_o$ denotes the estimated object boundary.

Reference-frame relevance. Directional relations require a frame. We assign higher scores to points whose perturbation changes the projection under π:

$$S_{frame}(p_i; r, \pi) = \left|u_\pi^\top (p_i - c_o)\right|, \tag{21}$$

where c_o is the object centroid.

4.5 Attack Objective

The total objective is

$$\min_\delta \ \mathcal{L}_{total} = \mathcal{L}_{rel} + \lambda_{sem}\mathcal{L}_{sem} + \lambda_{geo}\mathcal{L}_{geo} + \lambda_{smooth}\mathcal{L}_{smooth} + \lambda_{dens}\mathcal{L}_{dens} + \lambda_{phys}\mathcal{L}_{phys}. \tag{22}$$

subject to

$$\delta_i = 0 \text{ if } p_i \notin \mathcal{K}_{q,r}, \quad \|\delta_i\|_2 \le \varepsilon \text{ if } p_i \in \mathcal{K}_{q,r}. \tag{23}$$

Relation attack loss. If token logits are accessible,

$$\mathcal{L}_{rel} = -\log P_\theta(y_{tar} \mid \mathcal{P} + \delta, q). \tag{24}$$

If the answer is free-form, we map the answer to a normalized relation label through an answer normalizer g and use

$$\mathcal{L}_{rel} = \mathrm{CE}(g(F_\theta(\mathcal{P} + \delta, q)), r_{tar}). \tag{25}$$

For grounding, we additionally use

$$\mathcal{L}_{ground} = 1 - \mathrm{IoU}(B^{adv}, B_{tar}), \tag{26}$$

where B_{tar} is an incorrect but semantically plausible target box. For embodied affordance judgment, we use

$$\mathcal{L}_{aff} = \mathrm{CE}(a^{adv}, a_{tar}), \tag{27}$$

where a_{tar} is a target unsafe or incorrect action label.

Semantic preservation. We penalize changes to object identity:

$$\mathcal{L}_{sem} = \sum_{o\in\mathcal{O}_q} \mathrm{KL}\left(P(c_o \mid \mathcal{P}) \| P(c_o \mid \mathcal{P} + \delta)\right) + \mu_m\left(1 - \mathrm{IoU}(M_o, M_o^{adv})\right). \tag{28}$$

If class probabilities are unavailable, an external object validator is used and semantic preservation is measured as a hard validity condition.

Geometry. We use Chamfer and Hausdorff distances:

$$\mathrm{CD}(\mathcal{P}, \mathcal{P}^{\mathrm{adv}}) = \frac{1}{|\mathcal{P}|}\sum_{p\in\mathcal{P}} \min_{p'\in\mathcal{P}^{\mathrm{adv}}} \|p - p'\|_2^2 + \frac{1}{|\mathcal{P}^{\mathrm{adv}}|}\sum_{p'\in\mathcal{P}^{\mathrm{adv}}} \min_{p\in\mathcal{P}} \|p' - p\|_2^2, \tag{29}$$

$$\mathrm{HD}(\mathcal{P}, \mathcal{P}^{\mathrm{adv}}) = \max\{d_H(\mathcal{P}, \mathcal{P}^{\mathrm{adv}}), d_H(\mathcal{P}^{\mathrm{adv}}, \mathcal{P})\}, \tag{30}$$

$$d_H(A, B) = \max_{a\in A}\min_{b\in B} \|a - b\|_2. \tag{31}$$

with

$$\mathcal{L}_{geo} = \mathrm{CD}(\mathcal{P}, \mathcal{P}^{\mathrm{adv}}) + \rho\mathrm{HD}(\mathcal{P}, \mathcal{P}^{\mathrm{adv}}). \tag{32}$$

Local smoothness and density. To avoid isolated spikes,

$$\mathcal{L}_{smooth} = \sum_i \sum_{j\in\mathcal{N}(i)} w_{ij}\|\delta_i - \delta_j\|_2^2. \tag{33}$$

For density preservation, we voxelize the scene and penalize density shifts:

$$\mathcal{L}_{dens} = \sum_{v\in\mathcal{V}} |\rho_v(\mathcal{P}) - \rho_v(\mathcal{P}^{\mathrm{adv}})|. \tag{34}$$

Sensor plausibility. We restrict perturbations to surface-consistent displacements:

$$\mathcal{L}_{phys} = \sum_{i\in\mathcal{K}_{q,r}} \max(0, |n_i^\top \delta_i| - \varepsilon_n) + \lambda_{out}\mathbb{I}[p_i + \delta_i \notin \mathcal{S}_{vis}], \tag{35}$$

where n_i is a local surface normal and $\mathcal{S}_{vis}$ is a visible surface region estimated from camera or sensor pose when available.

4.6 Optimization

We use projected gradient descent over selected points:

$$\delta^{t+1} = \Pi_{\mathcal{C}}\left(\delta^t - \eta\nabla_\delta\mathcal{L}_{total}\right), \tag{36}$$

where $\Pi_{\mathcal{C}}$ projects perturbations into the allowed norm ball, relation-critical support set, and sensor-plausible surface constraint. If only point dropping is allowed, the optimization is replaced by a discrete ranking rule over $S(p_i; q, r)$. If only point addition is allowed, candidate points are sampled near relation-critical surfaces and optimized under density and outlier constraints.

Algorithm 1 White-box RelAttack3D

Require: Point cloud $\mathcal{P}$, query q, target wrong answer y_{tar}, model F_θ, budget ε, steps T.
Ensure: Adversarial point cloud $\mathcal{P}^{\text{adv}}$.

```
Parse q into relation graph G_q.
Extract relation tuples {τ_k} and involved objects O_q.
Localize object regions {Ω_o : o ∈ O_q}.
Compute relation-critical score S(p_i; q, r) for candidate points.
Select K_{q,r} ← TopK(S).
Initialize δ = 0.
for t = 1 to T do
    Construct P^adv = P + δ.
    Compute L_rel, L_sem, L_geo, L_smooth, L_dens, and L_phys.
    Update δ using projected gradient descent on K_{q,r}.
    Enforce δ_i = 0 for all p_i ∉ K_{q,r}.
    Project δ to norm, density, and sensor-plausibility constraints.
end for
return P^adv = P + δ.
```

4.7 Transfer and Query-Only Variants

For transfer attacks, we generate $\mathcal{P}^{\text{adv}}$ on a surrogate model F_ϕ and evaluate it on target model F_θ. To improve transferability, we optimize an ensemble objective:

$$\mathcal{L}_{ens} = \frac{1}{M}\sum_{m=1}^{M} \mathcal{L}_{rel}(F_{\phi_m}(\mathcal{P}+\delta, q), y_{tar}) + \lambda \mathcal{L}_{reg}. \quad (37)$$

We also randomize point sampling, rotations consistent with the reference frame, and density augmentations during optimization.

For query-only attacks, we estimate a relation score from model outputs. Let $s(\mathcal{P}, q, y_{tar})$ be a scalar score derived from target answer confidence, answer frequency across paraphrases, or an external relation judge. We update perturbations using natural evolution strategies:

$$\hat{\nabla}_\delta s = \frac{1}{m\sigma}\sum_{j=1}^{m} s(\mathcal{P}+\delta+\sigma u_j, q, y_{tar}) u_j, \quad (38)$$

where u_j are random directions restricted to $\mathcal{K}_{q,r}$. This preserves the core idea: even without gradients, the search space is reduced to relation-critical points.

4.8 Complexity

Let N be the number of points, k the number of relation-critical points, and T the number of optimization steps. Full-point PGD requires gradients over all N points. RelAttack3Dreduces the optimization support to $k \ll N$, lowering memory and improving interpretability. The additional cost comes from object localization and relation-critical scoring. When ground-truth masks are available, relation scoring is $O(N)$ plus nearest-neighbor computations for selected relation families. For distance and passability relations, nearest point pairs and corridor boundaries can be computed approximately with voxel grids or k-d trees.

Algorithm 2 Transfer / Query-Only RelAttack3D

Require: Target model F_θ, optional surrogates $\{F_{\phi_m}\}$, query budget Q.

```
Parse q and localize objects.
Mine K_{q,r} using surrogate gradients or geometry-only scoring.
if surrogates available then
    Optimize δ on surrogate ensemble.
    Evaluate F_θ(P + δ, q).
else
    for t = 1 to Q do
        Sample perturbation directions restricted to K_{q,r}.
        Query F_θ and estimate relation-score gradient.
        Update and project δ.
    end for
end if
return best adversarial point cloud under validity constraints.
```

5 RelBench-3D: Benchmark Construction

A strong paper cannot rely on hand-picked examples. We therefore define RelBench-3D, a relation robustness protocol constructed from existing 3D-language datasets.

5.1 Source Datasets

We recommend using the following sources:

- **ScanQA**: relation-rich 3D question answering over indoor scenes.
- **ScanRefer**: natural-language 3D grounding with object references.
- **SQA3D**: situated question answering with agent pose and viewpoint information.
- **ReferIt3D / Nr3D / Sr3D-style data**: fine-grained referring expressions with object disambiguation.
- **Embodied simulator subset**: optional passability and reachability questions built in Habitat, AI2-THOR, or an equivalent simulator.

5.2 Sample Filtering

Each candidate sample must pass four filters:

1. The query contains a supported relation from Table 1.
2. The involved objects can be localized by ground truth or a reliable detector.
3. The clean model answer is correct under the answer normalizer.
4. The target wrong answer is valid and non-trivial.

The third filter is important. If the clean model already answers incorrectly, the sample cannot measure adversarial vulnerability. It should be excluded from attack success statistics or counted separately as clean failure.

5.3 Target Answer Construction

For binary questions, the target answer is often the opposite label. For multi-choice or open-ended questions, the target answer must be semantically plausible. For grounding, the target box should be a distractor object that satisfies the same category when possible but violates the target relation. For embodied passability, the target action should be a safety-relevant opposite label such as passable versus blocked.

5.4 Answer Normalization

Free-form VLM outputs require robust normalization. We use a two-stage normalizer:

- A rule-based mapper handles yes/no, left/right, on/under, and common synonyms.
- A frozen language judge maps longer answers to relation labels, but only when confidence exceeds a threshold.

All ambiguous outputs are marked as invalid or separately reported. This avoids inflating attack success by exploiting answer-format noise.

5.5 Benchmark Splits

We recommend reporting results by relation family rather than only aggregate scores. A model may be robust to on/under but vulnerable to left/right. Aggregate attack success can hide these differences.

6 Relation-Consistency Verification Defense

A paper that only attacks may be viewed as incomplete. We therefore propose Relation-Consistency Verification (RCV), a lightweight defense and diagnostic baseline. RCV does not claim to solve adversarial robustness, but it exposes relation instability.

6.1 Logical Query Rewriting

For each query q, we generate logically equivalent or inverse-consistent queries:

$$\text{A left of B} \Leftrightarrow \text{B right of A}, \tag{39}$$

$$\text{A on B} \Leftrightarrow \text{B under A}, \tag{40}$$

$$\text{A closer to B than C} \Leftrightarrow \text{A farther from C than B}. \tag{41}$$

The language consistency score is

$$C_{lang} = \frac{1}{K} \sum_{k=1}^{K} \mathbb{I}[g(F(\mathcal{P}, q_k)) \equiv r_k]. \tag{42}$$

Low consistency indicates possible relation instability.

6.2 Explicit Geometric Checking

When object boxes or masks are available, we compute a geometric relation label:

$$r_{geo} = \psi(B_s, B_t, \pi), \tag{43}$$

where ψ is a deterministic geometric checker. For example, for left/right under frame π,

$$\psi(B_s, B_t, \pi) = \text{sign}(u_\pi^\top (c_s - c_t)). \tag{44}$$

A mismatch between r_{geo} and the VLM output r_{vlm} is treated as a warning.

6.3 Viewpoint Transformation Consistency

For viewpoint-dependent relations, we apply a known transformation T to the point cloud and transform the query accordingly:

$$F(T\mathcal{P}, Tq) \approx TF(\mathcal{P}, q). \tag{45}$$

An adversarial sample may break this equivariance even when the clean sample is stable.

6.4 Defense Score

We combine the three checks into a detection score:

$$D_{RCV} = w_1(1 - C_{lang}) + w_2 \mathbb{I}[r_{geo} \neq r_{vlm}] + w_3(1 - C_{view}). \tag{46}$$

A sample is flagged if $D_{RCV} > \tau_D$.

7 Experiments

This section is written in a submission-ready style, but numerical entries are placeholders. They must be replaced by real experiments.

7.1 Models

We recommend evaluating at least three model families:

- **Object-level point-language models**: PointLLM and GPT4Point-style models.
- **Scene-level 3D-language models**: 3D-LLM-style models supporting scene QA or grounding.
- **Open-vocabulary grounding models**: models that provide boxes or masks for object references.

If a model is not fully differentiable or not fully open source, it should be evaluated under transfer or query-only settings rather than silently treated as white-box.

Table 2: Recommended RelBench-3Dsplit structure. Values are placeholders and should be replaced after dataset construction.

Split	Directional	Support	Distance	Comparative	Topological	Affordance
ScanQA-derived VQA	TBD	TBD	TBD	TBD	TBD	–
ScanRefer-derived grounding	TBD	TBD	TBD	TBD	TBD	–
SQA3D situated QA	TBD	TBD	TBD	TBD	TBD	TBD
ReferIt3D referring expressions	TBD	TBD	TBD	TBD	TBD	–
Embodied simulator subset	TBD	TBD	TBD	–	TBD	TBD

7.2 Baselines

A convincing paper must compare against both generic attacks and relation-specific ablations:

- **Random noise**: uniformly perturb selected points.
- **Global PGD**: optimize all points without relation-critical mining.
- **Classification-critical attack**: select points using object classification saliency.
- **Point drop**: remove relation or classification critical points.
- **Point add**: add points near object boundaries under density constraints.
- **Caption attack**: optimize for a target caption without relation constraints.
- **Geometry-only attack**: use only relation geometry without VLM gradients.
- **RelAttack3D**: full query-conditioned relation-critical attack.

7.3 Metrics

We report the following metrics:

- **RASR**: relation attack success rate.
- **TASR**: targeted relation attack success rate.
- **VASR**: valid attack success rate, requiring relation error and semantic preservation.
- **SPR**: semantic preservation rate.
- **CD/HD**: Chamfer and Hausdorff distances.
- **Density shift**: voxel-wise density distribution change.
- **Outlier ratio**: fraction of points violating local density or surface consistency.
- **Grounding shift**: $1 - \mathrm{IoU}(B^{clean}, B^{adv})$.
- **Decision risk**: unsafe or wrong embodied action rate.
- **Query count**: number of target-model queries for black-box attacks.

The most important metric is VASR, not raw RASR. Raw RASR can be inflated by destroying object semantics; VASR cannot.

7.4 Main Results

7.5 Grounding and Captioning Results

7.6 Transfer and Black-Box Results

7.7 Ablations

The ablation study must prove that the method is not merely PGD with a new name.

7.8 Defense Evaluation

A useful defense result does not need to completely stop the attack. It should show whether relation inconsistency is detectable and where explicit geometry helps.

7.9 Human Imperceptibility Study

Because point-cloud perturbations are difficult to judge visually, we recommend a small human study. Participants see clean and adversarial point-cloud renderings from multiple views and answer whether they notice geometry corruption. Report detection accuracy and confidence. The study should not replace geometric metrics, but it strengthens the plausibility claim.

7.10 Qualitative Visualization

The final paper should include at least four visualization types:

- relation-critical heatmaps for each relation family;
- clean versus adversarial point-cloud comparisons;
- grounding shift examples;
- defense consistency examples showing equivalent-query contradictions.

8 Expected Analysis Structure

When real results are available, the analysis should avoid weak statements such as "our attack performs better." A top-conference analysis should answer the following:

Table 3: Main 3D VQA relational attack results. Replace all TBD entries with real measurements. VASR is the most important metric because it requires relation error and semantic preservation.

Method	RASR ↑	TASR ↑	VASR ↑	SPR ↑	CD ↓	HD ↓	Outlier ↓
Random noise	TBD	TBD	TBD	TBD	TBD	TBD	TBD
Global PGD	TBD	TBD	TBD	TBD	TBD	TBD	TBD
Classification-critical attack	TBD	TBD	TBD	TBD	TBD	TBD	TBD
Point drop	TBD	TBD	TBD	TBD	TBD	TBD	TBD
Caption attack	TBD	TBD	TBD	TBD	TBD	TBD	TBD
Geometry-only relation attack	TBD	TBD	TBD	TBD	TBD	TBD	TBD
RelAttack3D	TBD	TBD	TBD	TBD	TBD	TBD	TBD

Table 4: Relation-family analysis. This table is essential because aggregate attack success can hide which relations are truly vulnerable.

Relation Family	RASR ↑	VASR ↑	SPR ↑	CD ↓	Dominant Selected Points	Common Failure Mode
Directional	TBD	TBD	TBD	TBD	projected boundaries	viewpoint ambiguity
Contact/support	TBD	TBD	TBD	TBD	contact surface	support-plane confusion
Distance	TBD	TBD	TBD	TBD	nearest point pairs	nearest-neighbor flip
Containment	TBD	TBD	TBD	TBD	container boundary	inside/outside ambiguity
Comparative	TBD	TBD	TBD	TBD	order-sensitive points	ranking reversal
Topological	TBD	TBD	TBD	TBD	corridor boundary	adjacency confusion
Affordance/safety	TBD	TBD	TBD	TBD	clearance margin	unsafe passability answer

Does relation-critical mining matter? If full RelAttack3Doutperforms global PGD at similar CD and higher SPR, the result supports the central claim. If it only wins by using larger perturbations, the method is weak.

Which relations are most vulnerable? Directional and comparative relations may be more vulnerable because they depend on reference-frame projection and relative ordering. Contact/support relations may be more stable because they are constrained by vertical geometry. This should be tested, not assumed.

Does semantic preservation hold? A high raw RASR with low SPR is not a valid relation attack. The paper should emphasize VASR.

Does the attack transfer? If transfer is weak, the method may be overfitted to one model. If relation-critical geometry improves transfer, that is strong evidence that relation surfaces are shared vulnerabilities.

Can simple consistency checks detect the attack? If RCV detects many attacks, the paper can argue for geometry-language consistency as a promising defense. If not, the paper can argue that current 3D VLMs lack reliable self-consistency under spatial perturbation.

9 Discussion

9.1 Why This Is Not Just Another Point-Cloud Attack

The novelty of RelAttack3Dis not the use of PGD, Chamfer distance, or saliency. Those components are standard. The novelty lies in the attack target and the point-selection criterion. Existing point-cloud attacks usually optimize for object-level misclassification or perception failure. RelAttack3Doptimizes for relation-level failure under semantic preservation. This changes what counts as success, what points should be perturbed, what datasets should be used, and what metrics should be reported.

9.2 Failure Cases

We expect several failure cases:

- If a relation is determined by large-scale scene layout, small local perturbations may be insufficient.
- If the target model uses explicit geometric computation, relation hijacking may be harder.
- If object localization is unstable, it may be difficult to separate localization failure from relation reasoning failure.
- If the answer normalizer is noisy, attack success may be overestimated or underestimated.
- If directional labels lack a valid reference frame, the benchmark becomes ill-posed.

Table 5: Natural-language grounding attack results. The attack should cause grounding shift while preserving the semantic identity of the involved objects.

Method	Attack Success ↑	Grounding Shift ↑	Clean Object Preserved ↑	CD ↓	Valid Rate ↑
Random noise	TBD	TBD	TBD	TBD	TBD
Global PGD	TBD	TBD	TBD	TBD	TBD
Classification-critical attack	TBD	TBD	TBD	TBD	TBD
Caption attack	TBD	TBD	TBD	TBD	TBD
RelAttack3D	TBD	TBD	TBD	TBD	TBD

Table 6: Relation-aware captioning attack results.

Method	Err.↑	Obj.↑	Flu.↑
Global PGD	TBD	TBD	TBD
Caption attack	TBD	TBD	TBD
RelAttack3D	TBD	TBD	TBD

9.3 Limitations

This work is primarily a digital robustness study unless physical sensor experiments are added. A physically valid attack would need to demonstrate perturbations through RGB-D reconstruction noise, LiDAR artifacts, reflective surfaces, adversarial geometry, or small physical attachments. We also assume that object localization is available or can be estimated. In fully open-world scenes, object localization errors may dominate relation reasoning errors. Finally, many 3D VLMs are not fully open source, so white-box claims must be clearly separated from transfer and query-only claims.

9.4 Ethical Considerations

This work aims to identify and mitigate failures in 3D VLMs before they are deployed in robotics, AR/VR, assistive navigation, and embodied agents. Experiments should be conducted on public datasets and simulators. Released code should include evaluation scripts, defense baselines, and clear safety documentation. The paper should avoid instructions for causing real-world sensor harm and should emphasize robustness auditing.

10 Conclusion

We introduced RelAttack3D, a query-conditioned relational adversarial attack framework for point-cloud vision-language models. The main argument is that 3D VLM safety cannot be reduced to object recognition robustness or caption robustness. A model may preserve object semantics while failing at the spatial relations that drive grounding, manipulation, and navigation. RelAttack3Dexposes this vulnerability by parsing language queries into relation graphs, mining relation-critical points, and optimizing constrained perturbations that preserve semantics and point-cloud plausibility. We further proposed RelBench-3Dfor relation robustness evaluation and Relation-Consistency Verification as a lightweight diagnostic defense. The next step is empirical: implement the full pipeline, run experiments across multiple 3D VLMs and relation-rich datasets, and replace all placeholder results with reproducible measurements.

A Appendix A: Prompt Templates

A.1 VQA Templates

- Is the {subject} to the left of the {reference} from the agent's viewpoint?
- Is the {subject} on top of the {reference}?
- Which object is closer to the {reference}, the {candidate1} or the {candidate2}?
- Is the {subject} inside the {container}?
- Can the robot pass between the {object1} and the {object2}?

A.2 Grounding Templates

- Locate the {subject} to the left of the {reference}.
- Find the {subject} between the {reference1} and {reference2}.
- Select the object closest to the {reference}.
- Identify the object on top of the {reference}.

A.3 Defense Rewriting Templates

- A left of B $\Leftrightarrow$ B right of A.
- A on B $\Leftrightarrow$ B under A.
- A closer to B than C $\Leftrightarrow$ A farther from C than B.
- The gap between A and B is passable $\Leftrightarrow$ the robot can traverse between A and B without collision.

Table 7: Transfer attack results across model pairs. A top-conference paper should include transferability; otherwise the attack may be dismissed as white-box overfitting.

Source Model	Target Model	RASR ↑	VASR ↑	SPR ↑	CD ↓
PointLLM-style	GPT4Point-style	TBD	TBD	TBD	TBD
GPT4Point-style	PointLLM-style	TBD	TBD	TBD	TBD
3D-LLM-style	Point-language model	TBD	TBD	TBD	TBD
Ensemble surrogate	Held-out target	TBD	TBD	TBD	TBD

Table 8: Query-only attack efficiency.

Method	VASR ↑	Queries ↓	CD ↓
Random search	TBD	TBD	TBD
NES on all points	TBD	TBD	TBD
NES on relation-critical points	TBD	TBD	TBD
RelAttack3Dquery-only	TBD	TBD	TBD

B Appendix B: Geometric Checkers

For object boxes B_s and B_t with centroids c_s and c_t, a simple directional checker under axis u_π is

$$\psi_{left/right}(B_s, B_t, \pi) = \mathrm{sign}(u_\pi^\top (c_s - c_t)). \tag{47}$$

For support relation,

$$\begin{aligned}\psi_{on}(B_s, B_t) = \mathbb{I}[z_{min}(B_s) \geq z_{max}(B_t) - \tau_z] \\ \cdot \mathbb{I}[\mathrm{overlap}_{xy}(B_s, B_t) > \tau_o].\end{aligned} \tag{48}$$

For passability,

$$\psi_{pass}(B_a, B_b) = \mathbb{I}[\mathrm{clearance}(B_a, B_b) > w_{robot} + \tau_c]. \tag{49}$$

These checkers are not meant to replace VLM reasoning. They provide a consistency signal.

C Appendix C: Practical Implementation Checklist

- Build relation-filtered subsets and publish the filtering scripts.
- Report clean accuracy before attack.
- Exclude samples where the clean model already fails.
- Report VASR, not only RASR.
- Separate oracle localization from predicted localization.
- Include transfer and query-only settings.
- Include relation-family breakdowns.
- Visualize selected relation-critical points.
- Do not claim physical robustness or physical attack success unless tested with a sensor or simulator reconstruction pipeline.

Table 9: Ablation study. The key comparison is full RelAttack3Dversus variants without relation-critical mining and without semantic preservation.

Variant	RASR ↑	VASR ↑	SPR ↑	CD ↓	Outlier ↓	Interpretation
Full RelAttack3D	TBD	TBD	TBD	TBD	TBD	full model
Without gradient term	TBD	TBD	TBD	TBD	TBD	geometry only
Without relation geometry	TBD	TBD	TBD	TBD	TBD	gradient only
Without boundary term	TBD	TBD	TBD	TBD	TBD	no boundary prior
Without semantic preservation	TBD	TBD	TBD	TBD	TBD	may destroy objects
Without density/physical constraints	TBD	TBD	TBD	TBD	TBD	less plausible
Classification-critical points	TBD	TBD	TBD	TBD	TBD	wrong saliency target
Random selected points	TBD	TBD	TBD	TBD	TBD	no structure

Table 10: Relation-Consistency Verification defense.

Input Type	Detection AUC ↑	FPR ↓	Residual VASR ↓
Clean	TBD	TBD	–
Global PGD	TBD	TBD	TBD
Caption attack	TBD	TBD	TBD
RelAttack3D	TBD	TBD	TBD

Table 11: Human imperceptibility study template.

Method	Human Detect Acc. ↓	Confidence ↓	SPR ↑
Random noise	TBD	TBD	TBD
Global PGD	TBD	TBD	TBD
RelAttack3D	TBD	TBD	TBD

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